



NGUYEN BUI TAN DUNG

Entry-Level AI & Robotics Engineer | Applied AI Developer

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🌐 <https://github.com/tandung060604-prog> 🖥️ AI Demo: c2-app-028.up.railway.app

Professional Summary

Recent Robotics and Artificial Intelligence graduate with hands-on experience across **AI applications, computer vision, embedded systems, robotics control, backend APIs, and RAG/LLM products**. Built a physical AI-powered mobile robot using YOLOv8, Raspberry Pi 4, Arduino, PID control, camera-based localization, and a servo gripper. Also led an AI/RAG admissions assistant with FastAPI, PostgreSQL, citations, guardrails, handoff workflow, and evaluation metrics. Seeking entry-level roles in technology companies building AI products, automation systems, intelligent software, data-driven applications, or real-world AI deployment.

Core Technical Skills

Programming & Backend	Python, Arduino/C++, SQL, FastAPI, REST API design, PostgreSQL, Git/GitHub, Markdown technical documentation
AI / ML / Computer Vision	YOLOv8, CNN fundamentals, OpenCV, Roboflow, Google Colab, object detection, bounding-box processing, NMS, Precision/Recall/mAP evaluation
LLM / Applied AI	RAG, prompt engineering, citation-grounded answers, guardrails, refusal logic, human-in-the-loop handoff, golden dataset, evaluation checklist
Robotics & Embedded	Mobile robot, line following, PID control, Raspberry Pi 4, Arduino Nano, Serial UART, A4988 driver, TCRT5000 sensor, proximity sensor, servo gripper
Product & Collaboration	User stories, use cases, UI-API-RAG mapping, demo planning, project coordination, technical explanation for non-technical users

Selected Projects

AI-Powered Autonomous Mobile Robot for Object Detection & Transportation Mar 2026 - May 2026

Graduation Thesis Project | Tech: YOLOv8, Raspberry Pi 4, Arduino Nano, Python, OpenCV, Roboflow, TFLite/NCNN, PID, Serial UART

- Built an autonomous mobile robot that follows a line, detects target objects using camera-based AI, localizes object position, picks objects with a servo gripper, and transports them to target locations.
- Led the Computer Vision / Embedded AI part: YOLOv8 recognition pipeline, image processing, object localization logic, and Raspberry Pi-Arduino communication through Serial UART.
- Prepared and labeled image data on Roboflow, trained models on Google Colab, and compared YOLOv5, YOLOv8, and YOLOv11 under the same 100-epoch setup.
- Selected YOLOv8s for embedded deployment with Precision 0.993, Recall 0.995, and mAP@0.5 0.994; converted the model to TFLite/NCNN to improve inference on Raspberry Pi 4.
- Demo video: https://www.youtube.com/watch?v=_r1kk5_Yni0

HAUI Admission AI - Intelligent Admissions Assistant

Jun 2026 - Present

AI/RAG Lead & Project Lead | Tech: React/Vite, FastAPI, PostgreSQL, RAG, LLM, Guardrails, Evaluation

- Led AI/RAG and product-planning work for a Vietnamese admissions assistant that helps candidates and parents look up admissions methods, majors, benchmark scores, tuition, scholarships, documents, and FAQs.
- Designed RAG behavior for source-grounded answers with citations, academic-year filtering, clarification for ambiguous questions, safe refusal when sources are insufficient, and handoff tickets to human counselors.
- Prepared user stories, use cases, UI-API-RAG mapping, system prompt, golden dataset, evaluation checklist, and production smoke-test flow.
- Production demo: <https://c2-app-028.up.railway.app> | Demo credentials available upon request.

Industry, Research & Community Experience

Pegatron Vietnam - Equipment Operation Intern, AOI / Vision Inspection

Jan 2026 - Mar 2026

Cognex Vision Pro, AOI CCD1/CCD2 Caymus, Sony PS5 Controller Production

- Operated AOI CCD1/CCD2 Caymus equipment using Cognex Vision Pro in a consumer-electronics production line.
- Gained exposure to industrial visual inspection, defect checking, production-line constraints, equipment operation, and cross-cultural engineering workflows.

Closed-Chain Robot Research - Research Assistant / Mentor

Nov 2023 - May 2025

Robot kinematics, closed-chain mechanisms, finite symmetry theory

- Supported K15 and K16 students on kinematic-chain isomorphism checking and robot kinematics principles.
- Contributed to a closed-chain robot research project and a publication on a novel robot design using finite symmetry theory in the Journal of Science and Technology, HaUI.

STEAM for Vietnam - Volunteer / Project PM

Jul 2025 - Feb 2026

Robotics, electronics, and AI education

- Coordinated the BootUp Series and STEAMese Festival 2025 workshops in Hanoi and Ho Chi Minh City.
- Festival link: <https://www.steamforvietnam.org/steamese-festival2025>

Education & Achievements

Hanoi University of Industry (HaUI)

Graduated 2026

B.Eng. in Robotics and Artificial Intelligence | **GPA: 3.31/4.0** | **Top Graduate of the Major**

- Academic Encouragement Scholarship (3/7 semesters); Pegatron Talent Scholarship.
- Third Prize - Mitsubishi Electric Cup Automation Contest, university round (2024).
- Encouragement Prize - Student Scientific Research, School of Mechanical & Automotive Engineering (2025).

Leadership

Class Youth Union Secretary - Robotics & AI K17

Oct 2022 - Present

- Organized class activities, coordinated tasks, supported students, and strengthened communication, responsibility, project coordination, and teamwork skills.